

MAYUR VAIDYA

Assistant Professor (Feb 2020 – Present), DOB: 22nd Sept 1986

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1. Academic Qualification

	Degree	Graduation year	Subject	Institution	CGPA
1.	B. Tech+M.Tech	2010	Metallurgical & Materials Engineering	Indian Institute of Technology Madras	8.94/10
2.	PhD (Thesis: Thermal stability & Diffusion behaviour of CoCrFeNi and CoCrFeMnNi equiatomic high entropy alloys)	2017	Metallurgical & Materials Engineering	Indian Institute of Technology Madras	9.75/10

2. Professional experience

S.No.	Positions held	Name of the Institute	Duration
1.	Assistant Professor	Indian Institute of Technology Hyderabad	Feb 2020 - Present
2.	Post-Doctoral Researcher	Institute of Materials Physics, University of Muenster	May 2018 – Dec 2019
3.	Deputy Manager, R&D	ESSAR Steel Pvt. India Ltd., Hazira, Surat	July 2010 – July 2012

3. Area of expertise and Research Interests:

- Grain boundary diffusion in materials
- Interdiffusion in technologically important systems
- Oxidation behavior of metallic alloys
- Alloy development – Steel, High entropy alloys and light weight alloys
- Phase equilibria in multicomponent alloys

4. Sponsored Research Projects: (Total project cost: 3.4 Cr)

S.No	Title	Cost in lakhs	Role	Funding Agency	Start and End date
1	Atomic transport and phase growth in deformed transition metals	30	PI	SERB	22 nd Dec 2020 – 21 st Dec 2022
2	Using diffusion multiples to investigate interdiffusion in nanocrystalline materials produced by spark plasma sintering	25	PI	IIT Hyderabad (Institute Seed Grant)	15 th May 2020 – 15 th Aug 2022
3	Development of Borated Stainless Steel	25	PI	Volta Engg Design	1 st Nov 2022 – 31 st Dec 2023
4	Development of oxidation	1	PI	UGC-DAE	30 th March

	resistant nanocrystalline medium entropy alloys through diffusion analysis				2022 – 30 th March 2025
5.	Quantitative assessment of grain size and temperature effects on diffusion in medium entropy alloys	48	PI	ANRF	31 st May 2024 – 30 th May 2027
6.	Fabrication and characterization of High Entropy Shape Memory Alloys	45	PI	DYSL (DRDO)	8 th May 2024 – 31 st March 2026
7.	Thermo-kinetic Analysis of filler/Ni-based superalloy interface formed during vacuum brazing	53	PI	ARDB	Approved. Sanction order awaited.
8.	Computational assessment and development of new Ti-Al-Nb based alloys with rare earth metal additions for AM and PM applications	96	Co-PI	DST	20 Dec 2024 – 19 Dec 2027
9.	Microstructure analysis of steel specimens	17	Co-PI	Signode Pvt Ltd	Apr 2023 – Mar 2024
10.	Thermodynamic assessment towards the design of high temperature alloys	42	Co-PI	ANRF	Mar 2024 – Mar 2027

5. Awards and Honors

S. No	Name of Award	Awarding Agency	Year
1.	Top 2% Scientist – 2023, 2024. 2025	Stanford List	2023, 2024. 2025
2.	Editorial Board member	Scientific Reports Journal (Nature publications)	2025
3.	DAAD Research Ambassador for 2 consecutive terms	DAAD	2022-2025, 2025-2028
4.	Outstanding Materials Researcher	Venus International Foundation	2025
5.	Research Excellence Award – 2023 (below 40 years age category)	IIT Hyderabad	2023
6.	Teaching Excellence Award – 2023	IIT Hyderabad	2023
7.	INSA Young Scientist Award – 2022	Indian National Science Academy	2022
8.	Best paper award at In-house Symposium	Dept. of MME, IIT Madras	2017
9.	Best Publication award for the paper “J. Alloys Compd. 688 (2016) 994–1001”	Dept. of MME, IIT Madras	2016
10.	DAAD Sandwich Scholarship for PhD	DAAD	2014
11.	Institute Silver Medal for securing best academic record in Dual Degree	Dept. of MME, IIT Madras	2010
12.	DAAD WISE Scholarship for internship	DAAD	2009

6. Teaching (Total credits taught: 100, Average feedback: 4.5/5)

A. NPTEL Course

- (i) Diffusion in Solids - [Link](#)

B. Undergraduate courses

- (i) Phase Transformations
- (ii) Iron Making
- (iii) Diffusion in Solids
- (iv) Introduction to Materials Science and Engineering
- (v) Physical Metallurgy
- (vi) Phase Equilibria
- (vii) Metal Forming Laboratory

C. Post-graduate courses

- (i) Interdiffusion in Solids
- (ii) Advanced Phase Transformations
- (iii) Advanced Physical Metallurgy
- (iv) Advanced Thermodynamics of Materials
- (v) Diffusion Analysis in Materials Engineering
- (vi) High Entropy Materials

7. Professional Membership

- a) Indian Institute of Metals
- b) Powder Metallurgy Association of India

8. Research Supervision

- a) PhD Students: 10, Graduated - 1, Ongoing – 9
- b) M.Tech Students: 10: Graduated – 9, Ongoing – 1
- c) Summer Interns: 10

9. Research Outcomes (Total publications: 33, Total citations: 2100+, h-index: 17)

A. Publication List

1. B. Yadav, G. Mohan Murali Krishna, **M. Vaidya**, G. Wilde & S.V. Divinski. Grain boundary diffusion in compositionally complex alloys: A comprehensive review. **International Materials Reviews**, Accepted.

2. B. Yadav, T. N. Prasanthi, K. Guruvidyathri, M. Sadhasivam, C. Sudha, K. G. Pradeep, & **M. Vaidya** (2025). Grain boundary interdiffusion in CoFeNi/Sn system: Analysis of phase growth and compositional complexity. **Intermetallics**, 185, 108887.

3. N. K. Chaitanya, M. K. Singh, P. K. Ojha, S. Paul, S. Mallika, K. Guruvidyathri, P.P. Bhattacharjee & **M. Vaidya** (2025). Oxidation behavior of Al_{0.3}Cr_{1.3}Co₁Fe₁Mn₁Ni_{0.7} dual-phase high entropy alloy. **Journal of Alloys and Compounds**, 1022, 179860.

4. B. Yadav, A. Malgaonkar, & **M. Vaidya** (2025). Interdiffusion-Driven Phase Growth in Complex Concentrated FCC Alloys. **Transactions of the Indian Institute of Metals**, 78, 60.
5. S. M. Dikonda, A. Anupam & M. Vaidya, Oxidation Behavior of Mechanically Alloyed High-Entropy Alloys: A Review. **Advanced Engineering Materials**, 2401102.
6. S. Sen, M. Glienke, B. Yadav, **M. Vaidya**, K. Gururaj, K.G. Pradeep, L. Daum, B. Tas, L. Rogal, G. Wilde and S.V. Divinski (2024). Grain boundary self- and Mn impurity diffusion in equiatomic CoCrFeNi multi-principal element alloy. **Acta Materialia**, 264, 119588.
7. B. Yadav, A. Burla, J. Joardar, Guruvidyathri K., M. Sadhasivam, K. G. Pradeep & **M. Vaidya** (2024). Grain size effect on the phase growth in CoNi/Sn sandwich diffusion couples. **Materialia**, 33, 102011.
8. A. Anupam, K. Guruvidhyathri and **M. Vaidya** (2023), Development of high-entropy alloys as Bond Coats: A thermodynamic and kinetic perspective, **Transactions of Indian National Academy of Engineering**, 1-12S.
9. S. Das, M. Nagini, A. Anupam, K. Guruvidhyathri and **M. Vaidya** (2023). Oxidation behavior of $\text{CoCr}_{2-x}\text{FeNi}_{2.1}\text{Nb}_x$ high entropy alloys. **Journal of Alloys and Compounds**, 969, 172295.
10. N. K. Chaitanya, B. Yadav, P. P. Bhattacharjee, **M. Vaidya**, Effect of ultrafine microstructure on interdiffusion-driven phase transformations in Ni-Sn sandwich diffusion couples, **Materials Today Communications**, (2023): 105843.
11. B. Yadav, N. K. Chaitanya, M. Sadhasivam, J. Joardar, K. Guruvidyathri, K. G. Pradeep, **M. Vaidya**, Accelerated Phase Growth Kinetics During Interdiffusion of Ultrafine-grained Ni and Sn, **Journal of Alloys and Compounds**, 948 (2023) 169690.
12. A. Burla, M. Khandelwal, **M. Vaidya** Antibacterial properties of Cu containing complex concentrated alloys. **Materials Today Communications**, 33 (2022) 104915.
13. L.T. Belkacemi, **M. Vaidya**, S. Sevlikar, A. Hassanpour, F. Jomard, D. Irmer, C. Guerre, L. Martinelli, C. Duhamel, G. Wilde, V.A. Esin, S. V. Divinski, Intrinsic heterogeneity of grain boundary states in ultrafine-grained Ni: A cross-scale study by SIMS and radiotracer analyses. **Materialia** 22 (2022) p.101397.
14. A. Hassanpour, **M. Vaidya**, S.V. Divinski, G. Wilde, Impact of cryogenic cycling on tracer diffusion in plastically deformed Pd₄₀Ni₄₀P₂₀ bulk metallic glass, **Acta Materialia**, 209 (2021) 116785.
15. A. Parakh, **M. Vaidya**, N. Kumar, R. Chetty, B.S. Murty, Effect of crystal structure and grain size on corrosion properties of AlCoCrFeNi high entropy alloy, **Journal of Alloys and Compounds**, 863 (2021) 158056.
16. **M. Vaidya**, A. Karati, K. Guruvidyathri, M. Nagini, K.G. Pradeep, B.S. Murty, Suppression of σ -phase in nanocrystalline CoCrFeMnNiV high entropy alloy by unsolicited contamination during mechanical alloying and spark plasma sintering, **Materials Chemistry and Physics** 255 (2020) 123558.

17. **M. Vaidya**, Guruvidyathri K., B.S. Murty, Challenges in design and development of high entropy alloys: A thermodynamic and kinetic perspective, **Scripta Materialia**, 188 (2020) 37-43
18. **M. Vaidya**, Sandipan Sen, Lena Frommeyer, Lukas Rogal, S. Sankaran, Blazej Gabrowski, G. Wilde, S. V. Divinski, Phenomenon of ultra-fast tracer diffusion of Co in HCP high entropy alloys, **Acta Materialia**, 196 (2020) 220-230
19. M. Glienke, **M. Vaidya**, K.G. Pradeep, L. Daum, B. Tas, G. Wilde, S. V. Divinski, Grain boundary diffusion in CoCrFeMnNi high entropy alloy: Kinetic hints towards a phase decomposition, **Acta Materialia**, 195 (2020) 304-316
20. G. Mohan Muralikrishna, **M. Vaidya**, A. Karati, Soumya Ranjan Mishra, R. Bhattacharya, B.S. Murty, Influence of Al content on thermal stability of nanocrystalline $\text{Al}_x\text{CoCrFeNi}$ high entropy alloys at low and intermediate temperatures, **Advanced Powder Technology**, 31 (2020) 1985-1993
21. G. Mohan Muralikrishna, **M. Vaidya**, G. Wilde, B.S. Murty, S. V. Divinski, Tracer Diffusion in Ordered Pseudo-binary Multicomponent Aluminides, **Scripta Materialia**, 178 (2020) 227-231
22. R. John, A. Karati, M. Garlapati, **M. Vaidya**, R. Bhattacharya, D. Fabijanic, B.S. Murty, Influence of mechanically activated annealing on phase evolution in $\text{Al}_{0.3}\text{CoCrFeNi}$ high entropy alloy, **Journal of Material Science**, 54 (2019) 14588 - 14598
23. **M. Vaidya**, A. Anupam, V. Bhardwaj, C. Srivastava, B.S. Murty, Grain growth kinetics in CoCrFeNi and CoCrFeMnNi high entropy alloys processed by spark plasma sintering, **Journal of Alloys and Compounds**, 791 (2019) 1114-1121.
24. **M. Vaidya**, G. Mohan Muralikrishna, B.S. Murty, High entropy alloys by mechanical alloying: A review, **Journal of Materials Research**, 34 (2019) 664-686
25. **M. Vaidya**, Guruvidyathri K., B.S. Murty, Phase formation and thermal stability of CoCrFeNi and CoCrFeMnNi equiatomic high entropy alloys, **Journal of Alloys and Compounds**, 774 (2019) 856-864.
26. **M. Vaidya**, A. Karati, A. Marshal, K.G. Pradeep, B.S. Murty, Phase evolution and stability of nanocrystalline CoCrFeNi and CoCrFeMnNi high entropy alloys, **Journal of Alloys and Compounds**, 770 (2019) 1004-1015.
27. **M. Vaidya**, G. Mohan Muralikrishna, S. V. Divinski, B.S. Murty, Experimental assessment of the thermodynamic factor for diffusion in CoCrFeNi and CoCrFeMnNi high entropy alloys, **Scripta Materialia**, 157 (2018) 81-85.
28. **M. Vaidya**, K.G. Pradeep, B.S. Murty, G. Wilde, S. V. Divinski, Bulk tracer diffusion in CoCrFeNi and CoCrFeMnNi high entropy alloys, **Acta Materialia**, 146 (2018) 211-224.
29. A. Karati, **M. Vaidya**, B.S. Murty, Comparison of different processing routes for the synthesis of semiconducting AlSb, **Journal of Materials Engineering and Performance**, 27 (2018) 6196-6205.

30. M. Vaidya, K.G. Pradeep, B.S. Murty, G. Wilde, S.V. Divinski, Radioactive isotopes reveal a non-sluggish kinetics of grain boundary diffusion in high entropy alloys, **Scientific Reports** 7 (2017) 1–11.

31. M. Vaidya, A. Prasad, A. Parakh, B.S. Murty, Influence of sequence of elemental addition on phase evolution in nanocrystalline AlCoCrFeNi: Novel approach to alloy synthesis using mechanical alloying, **Materials Design**, 126 (2017) 37–46.

32. M. Vaidya, S. Trubel, B. S. Murty, G. Wilde, S. V. Divinski, Ni tracer diffusion in CoCrFeNi and CoCrFeMnNi high entropy alloys. **Journal of Alloys and Compounds**, 688 (2016) 994-1001.

33. M. Vaidya, S. Armugam, S. Kashyap, B. S. Murty, Amorphization in equiatomic high entropy alloys. **Journal of Non-Crystalline Solids**, 413 (2015) 8-14.

B. Conference Presentations (Invited)

1. M. Vaidya, Grain boundary diffusion in high entropy alloys, **Gordon Research Conference – Physical Metallurgy**, Boston, USA, 6th – 11th July 2025.

2. M. Vaidya, Oxidation behaviour of mechanically alloyed high entropy alloys, **PMAI PM Conference 2024**, 19th – 21st Feb 2025.

3. M. Vaidya, Diffusion controlled phenomenon in multicomponent alloys processed by mechanical alloying and spark plasma sintering, **PMAI PM Conference 2024**, 25th – 28th Feb 2024.

4. M. Vaidya, Interdiffusion driven phase transformations in multicomponent alloys, **METCENT 2023**, IIT BHU, 26th – 28th October 2023, Varanasi, India

5. M. Vaidya, Development of high entropy alloys for high temperature applications, **Structural Integrity Conference and Exhibition**, IIT Hyderabad, 14th – 16th December 2022, Hyderabad, India.

6. M. Vaidya, S. Sen, M. Glienke, M. M. Muralikrishna, J. Zhang, G. Laplanche, B.S. Murty, G. Wilde, S.V. Divinski “Diffusion behaviour of high entropy alloys with various solid solution structures”, **International conference on Diffusion in Solids and Liquids (DSL)**, 24th – 29th June 2019, Athens, Greece.

7. M. Vaidya, S. Sen, M. Glienke, D. Gaertner, G. Wilde, S.V. Divinski “Bulk and grain boundary diffusion in high entropy alloys”, **Annual Spring meeting of DPG**, 31st March – 5th March 2019, Regensburg, Germany.